

54 Community Ecology

KEY CONCEPTS

- 54.1** Interactions between species can help, harm, or have no effect on the individuals involved *p. 1215*
- 54.2** Diversity and trophic structure characterize biological communities *p. 1222*
- 54.3** Disturbance influences species diversity and composition *p. 1228*
- 54.4** Biogeographic factors affect community diversity *p. 1231*
- 54.5** Pathogens alter community structure locally and globally *p. 1234*

Study Tip

Make a table: As you read the chapter, build a table listing factors that can affect community structure. Add in figure numbers or page numbers that give examples of these factors.

Factor affecting community structure	What aspect of community structure is affected?	Example
Competition	Locations where a species can live in a community	Competition between two barnacle species (Figure 54.3)

Go to Mastering Biology

For Students (in eText and Study Area)

- Get Ready for Chapter 54
- Figure 54.17 Walkthrough: An Antarctic Marine Food Web
- HHMI Videos: Moth Mimicry: Using Ultrasound to Avoid Bats; Some Animals Are More Equal Than Others: Keystone Species and Trophic Cascades; Trophic Cascades in Salt Marsh Ecosystems; Riverine Food Webs: How Flow Rates Affect Biomass

For Instructors to Assign (in Item Library)

- Tutorial: Make Connections: Character Displacement and Modes of Selection
- Interpreting Data: Bird Populations During Old-Field Succession

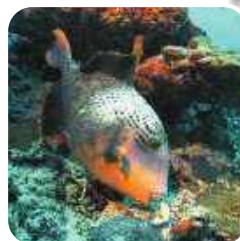


Figure 54.1 Although this moray eel could easily eat the cleaner wrasse in its mouth, the eel remains still, allowing the wrasse to eat parasites living in the eel's mouth. The wrasse and the eel—and the parasites in the eel's mouth—live together in a **community**, a group of populations of different species living in close enough proximity to interact.

What are some factors that influence the structure of a community?

Foundation species

Species that are large or abundant, such as the corals that form this reef, may affect community structure by providing habitat and food for other organisms, like the parrotfish below.



Interactions between species

Predation and other interactions affect the number of species in a community and the particular species that are present. Interactions can be classified in three categories:

- **Competition:** negatively affects both organisms involved
- **Exploitation:** benefits one organism while harming the other
- **Positive interactions:** benefit one or both organisms without harming either

Disturbances

Marine heat waves (resulting in coral bleaching, above), storms, human activities, and other disturbances can affect a community by removing organisms or altering resource availability.